

🔥 Fire-Fighting Robot - Complete System Architecture

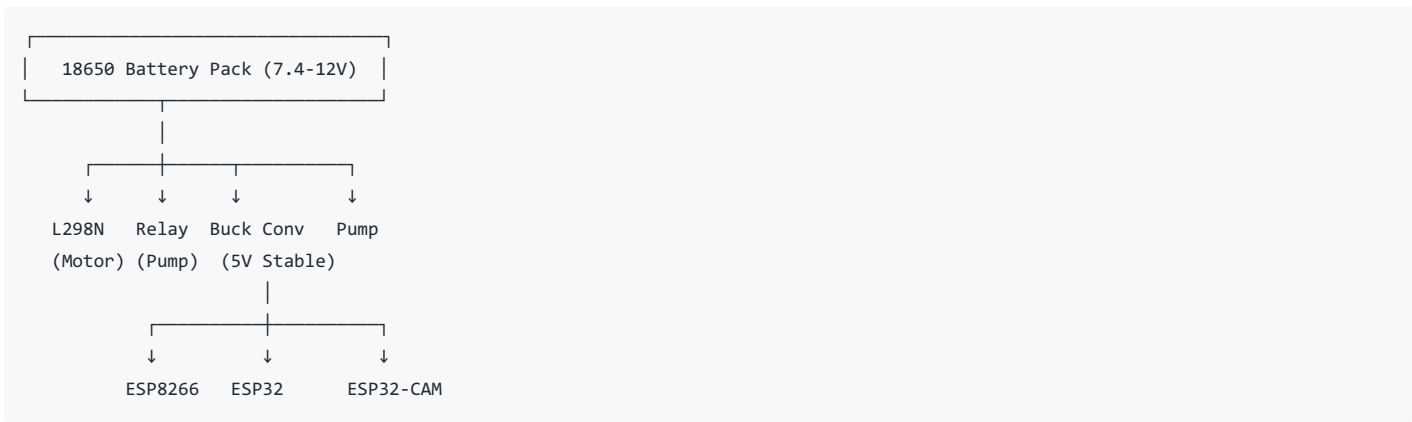
A fully autonomous fire detection and suppression system using multiple microcontroller boards

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⚡ Power Layout

System Power Distribution



⚠ Critical Rules

Rule	Status	Description
ALL GND Common	🔴 Required	Connect all ground lines together
ESP32-CAM → 5V Stable	🔴 Required	Use Buck Converter for stable power
No Direct Motor → ESP	🔴 Forbidden	DO NOT connect motor lines directly to microcontroller

🔧 ESP8266 → Motor Control (L298N)

Overview

Microcontroller: NodeMCU (ESP8266)

Function: Primary motor control and movement commands

Pin Configuration

ESP8266 Pin	GPIO	→	L298N Pin	Purpose
D5	GPIO14	→	ENA	Left Motor Enable
D6	GPIO12	→	ENB	Right Motor Enable
D8	GPIO15	→	IN1	Left Motor Forward

D7 ESP8266 Pin	GPIO13	→	IN3 L298N Pin	Left Motor Reverse Purpose
D4	GPIO2	→	IN3	Right Motor Forward
D3	GPIO0	→	IN4	Right Motor Reverse
GND	-	→	GND	Common Ground

Motor Connections

L298N Motor Outputs:

└ OUT1 + OUT2 → Left Motors

└ OUT3 + OUT4 → Right Motors

Power Supply

Battery (+12V) → L298N +12V

Battery (-GND) → L298N GND

Control Logic Example

Direction: FORWARD

└ Left: D8=HIGH, D7=LOW → OUT1,OUT2 Forward

└ Right: D4=HIGH, D3=LOW → OUT3,OUT4 Forward

Direction: REVERSE

└ Left: D8=LOW, D7=HIGH → OUT1,OUT2 Reverse

└ Right: D4=LOW, D3=HIGH → OUT3,OUT4 Reverse

Direction: LEFT TURN

└ Left: D8=LOW, D7=HIGH → Reverse

└ Right: D4=HIGH, D3=LOW → Forward

🔧 ESP32 → Sensors + Relay

Overview

Microcontroller: ESP32
Function: Fire detection, sensor monitoring, and pump activation

🔧 Flame Sensor

Flame Sensor	ESP32	Voltage	Purpose
VCC	3.3V	3.3V	Power Supply
GND	GND	-	Ground Reference
DO	GPIO34	Digital	Fire Detection Output

Logic: HIGH = Fire Detected

🔧 MQ-2 Gas Sensor

MQ-2 Sensor	ESP32	Voltage	Purpose
VCC	5V	5V	Power Supply
GND	GND	-	Ground Reference
DO	GPIO35	Digital	Gas Detection (Digital)
AO	GPIO34*	Analog	Gas Level (Analog, Optional)

*Requires voltage divider for safe analog reading

Logic: HIGH = Gas Detected

IR Sensor (Obstacle Detection)

IR Sensor	ESP32	Voltage	Purpose
VCC	3.3V / 5V	3.3V-5V	Power Supply
GND	GND	-	Ground Reference
OUT	GPIO27	Digital	Obstacle Detection

Logic: HIGH = Obstacle Detected

Relay + Pump

Control Side (Low Power)

Relay Module	ESP32	Purpose
VCC	5V	Power Supply
GND	GND	Ground Reference
IN	GPIO26	Activation Signal

Power Side (High Power)

Battery + (12V) → Relay COM (Common)
Relay NO (Normally Open) → Pump +
Pump - → Battery GND

Logic:

- GPIO26 = HIGH → Relay Energized → Pump ON
- GPIO26 = LOW → Relay De-energized → Pump OFF

ESP32-CAM → Camera Only

Pin Configuration

ESP32-CAM	Connection	Voltage
5V	Buck Converter Output	5V Stable

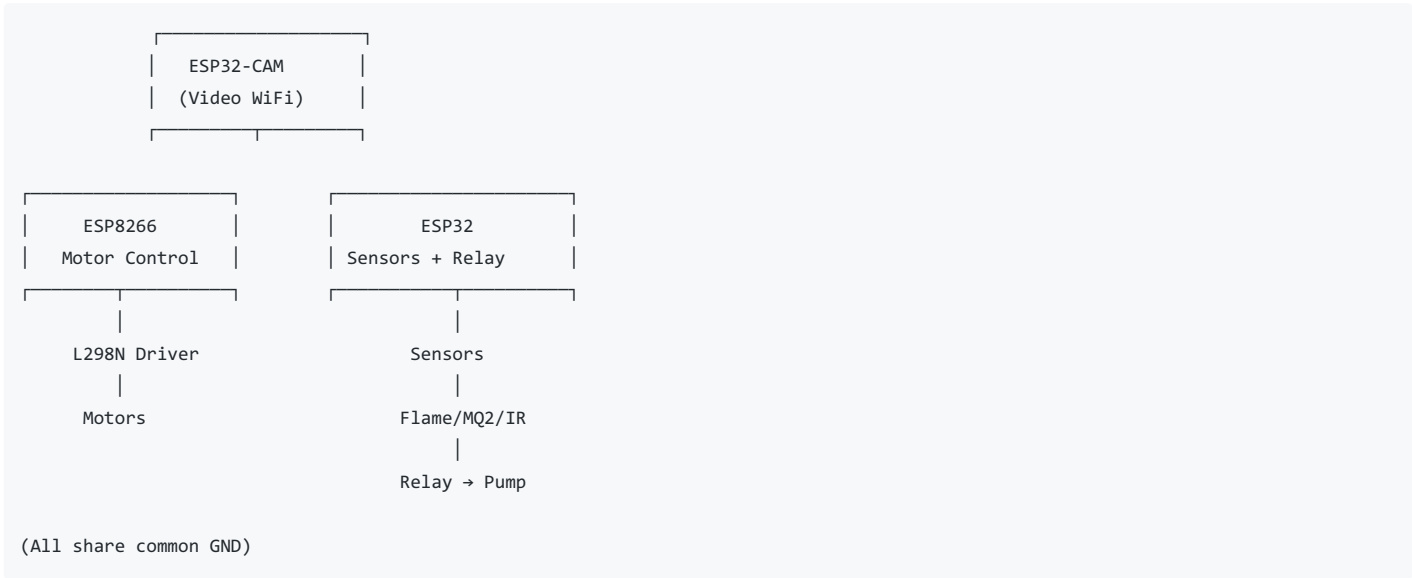
ESP32-CAM	Common GND	Voltage
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Key Specifications

- ❑ **Isolated System** - No connection to ESP32 or ESP8266
- ❑ **WiFi Streaming** - Direct WiFi video transmission
- ❑ **Independent Operation** - Self-contained video monitoring

Full System Integration

System Architecture Diagram



Device Responsibilities

Device	Role	Primary Function
ESP8266	Movement	Motor control & navigation
ESP32	Detection & Action	Fire detection & pump activation
ESP32-CAM	Observation	Real-time video streaming
Battery	Power	18650 pack (7.4-12V)

Working Logic

ESP8266 (Movement System)



🔧 ESP32 (Fire Detection System)



🔧 ESP32-CAM (Video Streaming)



📄 System Summary

Specifications at a Glance

Power Budget:	
└ Motor Power:	12V @ L298N (High Current)
└ Control Power:	5V Stable @ Buck Converter
└ Sensor Power:	3.3V / 5V Mixed
└ Camera Power:	5V Stable
Microcontrollers:	
└ ESP8266:	Motor Control (4 Motors)
└ ESP32:	Sensors (3 types) + Relay
└ ESP32-CAM:	Video Streaming
Sensors:	
└ Flame Detector	
└ Gas Sensor (MQ-2)	
└ Obstacle Sensor (IR)	
Actuators:	
└ 4x DC Motors (via L298N)	
└ Water Pump (via Relay)	